

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 07)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

* May use `main()` function approach (or multiple-functions) learnt in our earlier lesson in this lab

1. a) Write a Python program to create a list (named **Lab7_1.py**):
 - Create a list named `myList` and initialize it with 8 integer values below:
1, 3, 4, 0, -1, -12, -23, 289
 - Display them on a single line. (See sample output below.)
 - Determine and display the total size of the list by using built-in `len()`
(Hint: use **for loop** to display the values of the list.)

Sample Program Output

```
The values stored in myList are:  
1 3 4 0 -1 -12 -23 289  
The size of mylist is 8
```

2. Write a Python program to create a list. (named **Lab7_2.py**):
 - Create a list named **myList2** with the size of 6 integers.
 - 6 integers are input by user and stored in the **myList2**.
 - Display the input values on a single line.

(Hint: use **while** loop for user input and **for loop** for output display)

Sample Program Output

```
Enter 6 integers for the myList2:  
Enter # 1 : 4  
Enter # 2 : 0  
Enter # 3 : 20  
Enter # 4 : 123  
Enter # 5 : 321  
Enter # 6 : 111  
The values stored in myList2 are:  
4 0 20 123 321 111
```

3. Write a Python program (named **Lab7_3.py**) to let user enter a string and
- extract part of the string on screen (the first five characters), and display the result
 - Use slicing expression to select the first five characters (from 0 to 5) from the input string.
 - Display the selected string on screen
 - reverse the string when you print it on screen, and display the result
 - Check the size of the input string by `len()` and store it in a variable named `index`
 - Convert the string, with slicing in a reverse order.
 - Display the string in reverse order, (e.g. Hello => olleH).

Sample Program Output:

```
Enter a string: We all love ICP very much!!
First five characters are <We al>
Reverse String is <!!hcum yrev PCI evol lla eW>
```

4. Write a Python program (named **Lab7_4.py**) to ask user to input string and display each word from the string line by line.
- Write a **main()** function:
 - Ask user to input a string and accepts a string of characters in a variable named `message` in string type.
 - Call the function **dispStrWord()** to pass the input string (one word a line).
 - Write a programmer created function named `dispStrWord()` that receives a string argument
 - Display the input string, **one word per line**.

(Hint: may use `for` loop to assess the individual characters from the input string)

Important: type and execute the `main()` at the end of the program

Sample Program Output

```
Enter a string: We all love ICP very much!!
One word per line is:
We
all
love
ICP
very
much!!
```

~ END ~